

# Learned helplessness and mental health issues related to distance learning due to COVID-19

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## ABSTRACT

When the COVID-19 pandemic struck in March 2020, universities worldwide were forced to suddenly move all in-person students online. In isolation and away from their regular structures and coping mechanisms, students were forced to rely on online learning technology (OLT) as a full replacement for in-person learning. We hypothesize that students in this circumstance will experience feelings of learned helplessness regarding OLT and suffer from overall poorer mental health. We present a mixed-methods study to investigate these phenomena during the Spring 2020 semester among a diverse group of students. We explore multiple factors that contributed to these phenomena, such as motivation, growing exhaustion with online learning, and feelings of connectedness that were lost and cannot be recreated via online meeting software.

## CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

## KEYWORDS

learned helplessness, mental health, online learning, remote education, COVID-19

### ACM Reference Format:

Adam Garcia, Garrett Powell, Davis Arnold, Luis Ibarra, Matthew Pietrucha, Michael Thorson, Abigail Verhelle, Nathan Wade, and Samantha Webb. 2021. Learned helplessness and mental health issues related to distance learning due to COVID-19. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3451526>

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*CHI '21 Extended Abstracts*, May 8–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3451526>

## 1 INTRODUCTION

In the Spring of 2020, students were impacted in numerous ways by the COVID-19 pandemic. One of the most glaring impacts was on student learning. The sudden move to fully-online learning introduced students to several new learning technologies. The novelty of the scenario indicates that schools and universities have not had time to properly understand the impact on students forced into an all-online learning environment. With the pandemic still raging and many universities still fully online, this is an issue that will remain relevant for the foreseeable future.

With students required to use online learning technology (OLT) as a replacement for their on-campus experience, university officials predict in their planning that students will suffer from issues around “routine and self-discipline, mental/physical well-being, study motivation and feelings of isolation” [16]. Students also will be required to use technology to accomplish their schooling goals more than ever before and in new ways, which could create a feeling of learned helplessness in some students. Learned helplessness in the field of technology, as described by Don Norman, is a situation in which people experience repeated failure to use technology, perceive that others are not having the same problem, and begin to blame themselves [13]. With students in isolation and possibly suffering from exacerbated mental health issues, this scenario seems not only possible but highly likely.

In this paper, we present an empirical study designed to discover if students experienced learned helplessness and mental health issues due to OLT in the second half of the Spring 2020 semester. We conducted a two-part mixed-methods study consisting of a small set of pilot interviews to discover questions for a large-scale survey. In our analysis we answer the following research questions with regard to university undergraduate students who were primarily taking courses in a physical classroom:

- **RQ1:** To what extent did the “COVID semester” create feelings and behaviors of learned helplessness with regard to OLT?

- **RQ2:** To what extent did the "COVID semester" impact mental health?

**Contributions:** We provide two important contributions in this paper. Most importantly, this paper is the first to empirically investigate whether undergraduate students experienced learned helplessness with OLT during pandemic-related isolation. Second, we corroborate the early findings of others who found mental health issues connected with OLT. Both of these contributions are directly related to HCI as previous studies, such as those in the realm of usable privacy and security, have found that users experience learned helplessness when they feel a lack of control over their technology and these feelings are associated with mental health issues [19].

**Impact:** As demonstrated in Section 2, we contribute to a global conversation and provide critical data on how sudden emergency remote learning has impacted students so that universities around the world can better formulate plans for online learning for students who would otherwise be in a physical classroom.

## 2 BACKGROUND

Learned helplessness is a concept from psychology and medicine describing repeated failures at a task that have led to feelings of hopelessness and lack of control on one's environment [18]. This behavior has many implications for the HCI of distance learning. Thimbley reports that performance-oriented students who are accustomed to praise will develop learned helplessness when their work becomes too difficult, sabotaging themselves by shifting the blame to their own disorganization or some other non-personal factor (such as technology) [20]. Shklovski et al. discussed how their participants displayed learned helplessness, feeling trapped and hopeless by the lack of control of their data privacy in the modern world. [19]. Wang et al. reported that students who become frustrated with aspects of their learning experience that they cannot change can suffer from a lack of motivation leading to the development of learned helplessness [22].

Some information is beginning to appear with regard to the impact of the sudden shift to complete dependence on OLT and student experiences were mixed at best [8, 17]. As we face a pandemic, academics are struggling around the world to better understand its effects on university education. Crick et al. researched the point of view of computer science teachers in the UK toward the shift to online learning. They found that there was a positive attitude towards the shift with the expectation that it was hard to teach certain mathematical and programming concepts and skills to students in the online environment [6]. In Thailand, Pal et al. looked at how video conferencing applications (Zoom, Microsoft Teams, and Cisco Webex) would affect a student's ability to learn and which application students would be willing to use more often. They found that Zoom and Cisco Webex were most effective while Microsoft Teams was the least effective [14]. In China, Zhao et al. analyzed the importance of interactivity in education in the online environment between students, teachers, and the content. The results showed that in order for students to succeed in their studies the interactivity of students, teachers, and content must be maintained and strengthened when teaching online [23]. In Russia, a study was done by Kurbakova et al. to determine if the shift to online learning would still be able to provide the same learning process and flow.

After the study, the results showed that both the professors and students were content with the shift, and the same learning process and flow was still maintained [10]. Universities in countries without highly developed internet infrastructure, such as Pakistan, have had difficulties implementing widespread online education in place of residential programs [1].

Little is known about how the Spring 2020 semester impacted student mental health. Durham University, UK, predicted and planned for an increase in student mental health crises [16]. In a study with medical students in Saudi Arabia, students reported they had more time to focus and study, while also reporting that compared to before the pandemic they felt like they were now always busy, had no social life, and felt distracted by their learning environment [9]. Asanov et al. in Ecuador found that 16% of students in their sample reported depression related to social isolation [2]. Finally, a very recent study conducted at UC Berkeley examined the negative impacts on mental health that resulted because of the COVID-19 pandemic [5]. They reported a sharp rise in different types of mental health disorders and also discussed how the pandemic increased feelings of depression and anxiety in many students, but particularly those already vulnerable, such as LGBTQ students. Because of these studies, we expect to find mental health issues in our results as well, but we specifically examine whether those issues were related to the complete dependence on OLT to deliver the entire academic experience for students that were previously primarily taking face-to-face courses.

## 3 METHODOLOGY

We present a mixed-methods study using qualitative interviews to discover survey questions for quantitative analysis conducted in October 2020 at Abilene Christian University (ACU). Due to the unprecedented nature of the COVID-19 semester, many phenomena surrounding it are still unknown. Therefore, we chose to interview participants first to determine what questions we should be asking in the large-scale quantitative instrument.

For participation in our interviews or our survey, students got the opportunity to enter to win an assortment of prizes. The winners of the prizes filled out the proper tax forms to claim their prizes. Careful consideration was given to following all ethical guidelines under the supervision of ACU's institutional ethics review board, including in interview protocols, participant recruitment, and prize distribution. Every person in ACU's SIGCHI local chapter took the required ethical training given by the university. The training covers ethical issues based upon the protection of human subjects and the assurance on full disclosure including all risks and benefits to participation within our study.

### 3.1 Interviews

We used Zoom to conduct interviews to discover themes in student experiences during the target semester. Participants were specifically recruited with an eye toward diversity with regard to major, ethnicity, and gender. We conducted these interviews until we began to see similar themes recurring at a high rate. Each Zoom meeting consisted of the interviewee and two interviewers, one leading questions and the other a note-taker. The questions interviewers asked included their general opinion of online learning and

|     | Question                                                                                         | Mean | SD   |
|-----|--------------------------------------------------------------------------------------------------|------|------|
| 1.  | I feel like my workload increased after the switch to online learning.                           | 3.34 | 1.04 |
| 2.  | I felt more distracted while trying to learn online than I did in a physical classroom.          | 4.46 | 0.89 |
| 3.  | My technology was sufficient and reliable for learning online.                                   | 4.2  | 0.86 |
| 4.  | I looked at screens less in my free time because of the time I spent looking at them for school. | 2.3  | 1.23 |
| 5.  | Online learning negatively impacted my grades (including taking a no-credit in a course).        | 3.15 | 1.31 |
| 6.  | I felt less stressed and/or anxious about my coursework after the switch to online learning.     | 2.65 | 1.29 |
| 7.  | I felt lonelier after the switch to online learning.                                             | 4.18 | 1.14 |
| 8.  | The interaction available to me through online class was enough to keep me feeling connected.    | 2.16 | 0.98 |
| 9.  | Zoom, Google Meet, and Canvas were adequate for what I needed to learn.                          | 3.45 | 1.06 |
| 10. | I was more tired and fatigued after the switch to online learning.                               | 3.72 | 1.26 |
| 11. | I felt less motivated to do my work when it was all online.                                      | 4.36 | 1.04 |
| 12. | It was easy to adjust to all online learning.                                                    | 2.89 | 1.07 |
| 13. | Over the course of the semester, I felt like I had improved my ability to learn online.          | 3.53 | 1.05 |
| 14. | I got tired of trying to use OLT.                                                                | 3.63 | 1.22 |
| 15. | I felt like many of my problems with online learning were due to my ability to use technology.   | 1.96 | 1.04 |

**Table 1: Results of the quantitative survey with mean and standard deviation.**

the effect on their mental health and overall education. We allowed space in each interview to probe deeper into a student's experience if a particular question hit on something specifically interesting or relevant to the student.

Once all interviews were completed, we used an open-coding qualitative framework to tag the interview transcriptions for themes. All of these first-order themes for each user were put into a spreadsheet, with each user being a single column. We then collaborated as a group on merging the first-order themes into groups to create second-order themes. This interview process allowed for the discovery of important topics to ask about in our general survey.

### 3.2 Survey

After the intensive interview process, we were able to craft questions for our survey that were more specific to actual student experiences, while also remaining broad enough to be easily answered. For the survey, we used a five-point Likert scale, where 1 represented "strongly disagree" and 5 represented "strongly agree." See Table 1 below for a list of the questions as well as the mean and standard deviation of student responses. Each question was intended to cover one second-order theme that we discovered during our interview process. Some second-order themes, such as mental health, needed more than one question to adequately cover the complex dimensions of the topic. We alternated questions framed positively and negatively to balance the survey and be able to more easily detect someone running down the list with all ones or all fives.

The survey consisted of three parts. The first contained the consent form. The second contained three demographics questions about student year (first, second, third, fourth, fifth plus), student gender (male, female, non-binary, prefer not to say, prefer to self-identify), and student major. The third contained 15 research questions developed from the qualitative interview and coding process described above. Participants were recruited by the authors circulating the survey among undergraduate college students, direct appeals from various professors to their classes, and inclusion in

several department newsletters. To encourage participants to fill out the survey, prizes were offered to the participants.

## 4 RESULTS

In this section, we present our interview findings, how that led to the creation of the survey, and the results of the survey.

### 4.1 Interviews

We interviewed eight students. Researchers carefully watched the recorded interviews and reviewed the notes taken by the initial interview team. From each review, researchers noted the themes that emerged and put them into a spreadsheet. After themes had been recorded for all eight interviews, all researchers met together and reviewed them, coming to a consensus on 98 total first-order themes. Finally, all researchers grouped these themes into second-order themes in a collaborative and iterative process. After initially finding 15 second-order themes, some were grouped together as they were similar (e.g. themes relating to mental health), leaving a final total of 12 second-order themes. The second-order themes were: More Work, Distractions, Tech Obstacles, Tech for Recreation, Grades, Mental Health, Connection, Online Learning Tools (Zoom, Lockdown, Canvas), Physical Condition, Motivation/Procrastination, Adjusting, and Overall School Comments. We then used these 12 second-order themes to create our survey questions. While our initial goal was to have one question per second-order theme, some of the more complex themes, such as students' relationship with and feelings about OLT received more than one question.

### 4.2 Survey

There was a total of 115 respondents to our survey. Nine of those were first-year students who were not at ACU (i.e. in high school) during the "COVID semester" and were thus removed from the sample, leaving 106 respondents. Of those respondents, 41.5% (44) were male and 58.5% (62) female. Although our demographic questions

| Factor # | Factor Name          | Dependent Variable  | Independent Variable                                   | p-value                            | F-value               |
|----------|----------------------|---------------------|--------------------------------------------------------|------------------------------------|-----------------------|
| 1        | Use of Technology    | Tired of Using OLT  | Physically Tired<br>Felt Lonely<br>Self Blame with OLT | p < 0.001<br>p < 0.005<br>p < 0.07 | 18.67<br>8.52<br>3.57 |
| 2        | Motivation           | Less Motivated      | More Distracted<br>Felt Lonely                         | p < 0.001<br>p < 0.001             | 35.58<br>16.85        |
| 3        | Available Software   | Felt OLT Adequate   | Improved Ability to Learn Online                       | p < 0.001                          | 16.70                 |
| 4        | Personal Connections | Felt Connected      | Less Stressed                                          | p < 0.004                          | 9.23                  |
| 5        | Gender Effect        | Self Blame with OLT | Gender Identity                                        | p < 0.002                          | 10.43                 |
| 6        | Grades               | Grades Hurt         | More Distracted<br>Sufficient Technology               | p < 0.004<br>p < 0.04              | 8.73<br>4.52          |

**Table 2: Results of the exploratory factor analysis with six factors and the multivariate ANOVA tests run on each factor where the dependent variable in each factor had the highest factor loadings.**

included the options “non-binary”, “prefer not to answer”, and “prefer to self-describe”, none of those were selected by any respondents and we discuss this further in Section 6 Limitations. The sample includes a diverse set of majors, including Business (24.5%), Computer Science (15.1%), Nursing (10.4%), Game Development (8.5%), Education (6.6%), Biology (4.7%), Psychology (3.8%), Physics (3.8%), and 13 other majors with 3 or less respondents (22.6%). Finally, the sample also included a good balance of student year, with 39.6% second-year, 29.2% third-year, 26.4% fourth-year, and 4.7% fifth-year students. Response data to the questions are listed in Table 1 and we discuss these results and their implications in Section 5 Discussion below.

## 5 DISCUSSION

We next examine this question through the lens of our two research questions about learned helplessness with OLT and mental health.

In order to see causal relationships between phenomena, we first created a correlation matrix for all fifteen questions of our survey. This showed that, despite writing the questions from the themes that arose from the interviews and trying to aim them towards our two research questions, several of the survey questions were independent. Therefore, because it would be more difficult to find relationships in the data, we next ran an exploratory factor analysis with varimax rotation. We examined the fit of the model with three, four, five, six, and seven factors. The six-factor analysis provided the best fit for the data where all loadings totals for each factor were above 1.0, the cumulative variance was 46%, and  $X^2(49, N=106) = 50.1, p = 0.429$ . In this case, a high p-value is desired as that leads us to reject the null hypothesis (i.e. that six factors are not sufficient). In a meta-analysis of exploratory factor analysis and factor loadings, Peterson found that the average cumulative variance accounted for was 56.6% [15]. Tinsley and Tinsley reported that in social sciences, the cumulative variance is often less than 50% [21]. Each of the six factors are presented in Table 2.

### 5.1 Factors Relating to Learned Helplessness

Factors 1, 3, and 5 were about themes related to learned helplessness with regard to OLT. Our ANOVA tests for these factors (see Table 2) shows causal relationships that contributed to students’ feelings about OLT.

**Use of Technology** (Factor 1) - Experts predicted that students would suffer from lack of routine and physical exhaustion [16]. Our analysis highlights how physical exhaustion and loneliness from isolation at home can contribute to a general feeling of mental exhaustion with using OLT with a correlation to blaming OLT. In our interviews, we found that students fell out of their routines, such as not being required to attend an 8:00 am class because it’s been totally placed online, and did not plan adequately for sleep. They also reported they missed seeing their friends in classes and extracurricular activities.

**Available Software** (Factor 3) - Previous work found that applications such as Zoom were best for student engagement and interactivity [23]. We found that feeling that online learning software, such as Canvas and Zoom, are adequate means by which to take a college course directly contributes to a feeling of success in online learning. As in [9], we found that students who already enjoy online learning, can keep a good sleep schedule, and don’t feel isolated are less likely to blame their problems on their own ability to use technology. Certainly, no one expected to be totally dependent on OLT when the semester began and the sudden shift may have created a situation in which it became easy to get tired of OLT.

**Gender Effect** (Factor 5) - Even though the student population as a whole was not very likely to experience learned helplessness, we found that among those who did, women were significantly more likely to blame their problems with OLT on their ability to use said technology. For question 15 of our survey, the mean response for men was 1.59 (SD=0.90) and women had a mean of 2.21 (SD=1.06), a difference in means of 0.62,  $t(105) = 3.133, p < 0.025$ . While men are not better at using technology than women [12], our finding here is probably related to the well-known phenomenon of

lower self-efficacy in women than in men with regard to the use of technology [3, 11].

## 5.2 Factors Relating to Mental Health

Factors 2, 4, and 6 were centered on themes of mental health. The ANOVA tests for these three factors (see Table 2) shows the relationships between different variables regarding students' evaluation of their mental health during a time in the semester when they could only experience school through OLT.

**Motivation** (Factor 2) - Previous work found that students in pandemic-related isolation reported being more distracted by being at home than when they were at school [9]. We found that high levels of distraction, as well as feelings of loneliness, impact students' ability to stay motivated in their schooling. From the interviews we conducted, we discovered that students felt more distracted by family members, availability of entertainment that would not be available if they were physically in class, and a lack of structure.

**Personal Connectedness** (Factor 4) - Isolation has been found to be a large contributing factor to mental health-related issues during the pandemic for undergraduate students [5]. We found that students who felt less stressed retained a feeling of connectedness. Some of the interviewed students reported using online communication tools to speak with friends more often and searched for ways to simulate a classroom environment. The majority of students responding to the survey felt that the interaction available to them during the online-learning-only period of the semester was inadequate for them to feel connected. This lack of proper connection, coupled with other stresses and difficulties of online learning, may lead to increased stress and a decrease in overall mental health.

**Grades** (Factor 6) - Our analysis emphasizes that students who felt that their grades were negatively affected by the switch to fully-online learning felt more distracted by their environment. We grouped this factor into those pertaining to mental health because distraction appeared also in Factor 2 and is therefore related to issues of loneliness and motivation. However, these same students who felt their grades were negatively impacted also tended to feel that their technology was sufficient for their online learning experience. This was very revealing to us in that students' negative feelings about the most tangible outcome of the COVID-19 semester (i.e. their grades) centered around issues related to mental health and not about the sufficiency of their technology.

## 6 LIMITATIONS

There were several limitations to our study. First, the sample size for the interviews was small, but this is to be expected for exploratory interviews. It is important to note that no conclusions were drawn from these interviews, but rather they were used to help elicit themes from the students themselves so that the researchers would have a better idea of what questions to ask in the large-scale survey. Second, the sample size for the survey was small but was still able to yield interesting and statistically significant results. Furthermore, garnering large response rates during socially-distanced education, even online, is challenging. Third, there were no LGBTQ+ respondents in the survey, potentially biasing its findings in areas such as mental health, which is typically more at-risk in LGBTQ+ individuals, especially during the pandemic [5]. Fourth, taking the

sample at just one university could definitely bias the results, perhaps most evident in the case of the absent LGBTQ+ respondents, but could have produced other biases of which the researchers are not aware. Finally, exploratory interviews and surveys do not have the most explanatory power for human behavior, but we chose these research methods due to ACU's restrictions on most other research with human participants during the COVID-19 pandemic.

## 7 CONCLUSION

In this paper, we presented a mixed-methods study to discover if students developed learned helplessness and mental health issues related to OLT during the second half of the Spring 2020 semester due to the COVID-19 pandemic. The results and discussion presented above allow us to answer our research questions. In response to RQ1, we found that relatively few students outright experienced learned helplessness with OLT. However, many students reported experiences that could contribute to learned helplessness as shown in our factor analysis above. In response to RQ2, we found that many students struggled with mental health due to their isolation and their environment after the rapid shift from in-person to fully-online learning. Based on our interviews and survey analysis, we believe that courses which were designed to be more interactive and engaging were more successful at supporting the mental health of their students. Recent research has found that professors that engaged in promoting those kinds of interactions in their courses were more successful [4, 7]. This study should help universities around the world better understand the effects of OLT on their student body and make adjustments to enable their students to be more successful.

There is still much to be done in future work. First, a larger, multi-national, multi-institutional study is needed to determine more robust results. Second, our results indicate that students still struggled with their grades due to their mental health and learning environment, despite largely feeling comfortable with OLT. A follow-up study to more closely investigate this phenomenon would be beneficial to institutions of higher education. Third, our results may also be relevant to K-12 schools, many of which have gone fully online for part or all of the current school year, though obvious differences exist between university students and K-12 students. Finally, our results agree with prior work that interactivity and community in online class formats are key to student engagement. It would be beneficial to study the methods instructors used to make their courses more interactive while online and how those strategies interacted with the factors observed in our study.

## ACKNOWLEDGMENTS

The authors would like to thank Adam Saxton, who contributed to this work before his graduation, and Dr. James Prather, the faculty sponsor for the ACU SIGCHI Local Chapter, for his mentoring, help, and support. Thank you to Dr. John Homer, Dr. Brent Reeves, Dr. Brett Becker, and Dr. Paul Denny for their feedback on a draft of this paper. We would also like to thank Abilene Christian University, which supports our SIGCHI Local Chapter. Finally, we would like to thank the CHI reviewers who provided feedback before publication.

## REFERENCES

- [1] Muhammad Adnan and Kainat Anwar. 2020. Online Learning amid the COVID-19 Pandemic: Students' Perspectives. *Online Submission* 2, 1 (2020), 45–51.
- [2] Igor Asanov, Francisco Flores, David McKenzie, Mona Mensmann, and Mathis Schulte. 2020. Remote-learning, time-use, and mental health of Ecuadorian high-school students during the COVID-19 quarantine.
- [3] Zhihui Cai, Xitao Fan, and Jianxia Du. 2017. Gender and attitudes toward technology use: A meta-analysis. *Computers & Education* 105 (2017), 1–13.
- [4] Richard Allen Carter Jr, Mary Rice, Sohyun Yang, and Haidee A Jackson. 2020. Self-regulated learning in online learning environments: strategies for remote learning. *Information and Learning Sciences* (2020).
- [5] Igor Chirikov, Krista M Soria, Bonnie Horgos, and Daniel Jones-White. 2020. Undergraduate and Graduate Students' Mental Health During the COVID-19 Pandemic. (2020).
- [6] Tom Crick, Cathryn Knight, Richard Watermeyer, and Janet Goodall. 2020. The Impact of COVID-19 and "Emergency Remote Teaching" on the UK Computer Science Education Community. In *United Kingdom & Ireland Computing Education Research conference*. 31–37.
- [7] Alanna Gillis and Laura M Krull. 2020. <? covid19?> COVID-19 Remote Learning Transition in Spring 2020: Class Structures, Student Perceptions, and Inequality in College Courses. *Teaching Sociology* 48, 4 (2020), 283–299.
- [8] Tawnell D Hobbs and Lee Hawkins. 2020. The results are in for remote learning: It didn't work. *Wall Street Journal* 5 (2020).
- [9] Rehana Khalil, Ali E Mansour, Walaa A Fadda, Khaled Almisnid, Mohammed Aldamegh, Abdullah Al-Nafeesah, Azzam Alkhalifah, and Osama Al-Wutayd. 2020. The sudden transition to synchronized online learning during the COVID-19 pandemic in Saudi Arabia: a qualitative study exploring medical students' perspectives. *BMC medical education* 20, 1 (2020), 1–10.
- [10] Svetlana Kurbakova, Zlata Volkova, and Alexander Kurbakov. 2020. Virtual Learning and Educational Environment: New Opportunities and Challenges under the COVID-19 Pandemic. In *2020 The 4th International Conference on Education and Multimedia Technology*. 167–171.
- [11] Alex Lishinski, Aman Yadav, Jon Good, and Richard Enbody. 2016. Learning to program: Gender differences and interactive effects of students' motivation, goals, and self-efficacy on performance. In *Proceedings of the 2016 ACM Conference on International Computing Education Research*. 211–220.
- [12] Laurie Murphy, Brad Richards, Renée McCauley, Briana B Morrison, Suzanne Westbrook, and Timothy Fossum. 2006. Women catch up: gender differences in learning programming concepts. *ACM SIGCSE Bulletin* 38, 1 (2006), 17–21.
- [13] Don Norman. 2013. *The design of everyday things: Revised and expanded edition*. Basic books.
- [14] Debajyoti Pal, Vajirasak Vanijja, and Syamal Patra. 2020. Online Learning During COVID-19: Students' Perception of Multimedia Quality. In *Proceedings of the 11th International Conference on Advances in Information Technology*. 1–6.
- [15] Robert A Peterson. 2000. A meta-analysis of variance accounted for and factor loadings in exploratory factor analysis. *Marketing letters* 11, 3 (2000), 261–275.
- [16] Rille Raaper and Chris Brown. 2020. The Covid-19 pandemic and the dissolution of the university campus: Implications for student support practice. *Journal of Professional Capital and Community* (2020).
- [17] Maila DH Rahiem. 2020. The emergency remote learning experience of university students in indonesia amidst the COVID-19 crisis. *International Journal of Learning, Teaching and Educational Research* 19, 6 (2020), 1–26.
- [18] Martin EP Seligman. 1972. Learned helplessness. *Annual review of medicine* 23, 1 (1972), 407–412.
- [19] Irina Shklovski, Scott D Mainwaring, Halla Hrunnd Skúladóttir, and Höskuldur Borghorsson. 2014. Leakiness and creepiness in app space: Perceptions of privacy and mobile app use. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2347–2356.
- [20] Harold Thimbleby. 2009. Teaching and learning HCI. In *International Conference on Universal Access in Human-Computer Interaction*. Springer, 625–635.
- [21] Howard E Tinsley and Diane J Tinsley. 1987. Uses of factor analysis in counseling psychology research. *Journal of counseling psychology* 34, 4 (1987), 414.
- [22] Ruijie Wang, Liming Chen, Ivar Solheim, Trenton Schulz, and Aladdin Ayes. 2017. Conceptual motivation modeling for students with dyslexia for enhanced assistive learning. In *Proceedings of the 2017 ACM Workshop on Intelligent Interfaces for Ubiquitous and Smart Learning*. 11–18.
- [23] Rong Zhao, Xinghuan Luo, and Yumeng Chen. 2020. The Essence of Interactivity and Its Impact on Online Learning Community. In *Proceedings of the 2020 The 6th International Conference on Frontiers of Educational Technologies*. 6–10.